

ERS

ENGINEERING READINESS ASSESSMENT

INT8 Post-Training Quantization for Seq2Point NILM on Resource-Constrained Edge Platforms

ERS-2026-001

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Radianode Ltd

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SUPPORTING ARTEFACTS github.com/radianode/ers-2026-001	

EXECUTIVE SUMMARY

This Engineering Readiness Assessment evaluates whether INT8 post-training quantization provides a deployment-ready optimisation for a Seq2Point Non-Intrusive Load Monitoring (NILM) model intended for resource-constrained Edge AI platforms.

The investigation was conducted using the Radianode engineering validation methodology, focusing on the engineering characteristics that determine deployment readiness rather than theoretical model performance.

The measured evidence demonstrates that the evaluated optimisation:

- Reduced model parameter storage by 75%
- Preserved FP32 prediction behaviour
- Produced no material degradation in measured inference accuracy relative to the FP32 reference
- Introduced no observed deployment-critical numerical instability within the defined evaluation scope

Based on the engineering evidence presented in this assessment, the measured results support the deployment-readiness decision for the evaluated model within the defined scope.

ENGINEERING DECISION

Can INT8 post-training quantization be adopted as a deployment optimisation for the evaluated Seq2Point NILM model without compromising engineering confidence?

This publication answers that question using measured engineering evidence.

INVESTIGATION SCOPE

ENGINEERING OBJECTIVE

Evaluate whether INT8 post-training quantization preserves deployment readiness of the evaluated Seq2Point NILM model.

ENGINEERING ASSET	Seq2Point CNN NILM model
TARGET APPLIANCE	Water Heater (Boiler)
OPTIMISATION EVALUATED	INT8 post-training quantization
DEPLOYMENT SCENARIO	Resource-constrained Edge AI
EVALUATION DATASET	UK-DALE
MODEL ARCHITECTURE	Seq2Point CNN
IMPLEMENTATION FRAMEWORK	MATLAB Deep Learning Toolbox

Target-appliance context. The evaluated Seq2Point CNN is configured for target-appliance disaggregation of the Water Heater (Boiler) from aggregate household electricity consumption. The FP32 and INT8 models therefore produce estimates of the same target-appliance signal, and the reported inference comparisons evaluate the effect of INT8 post-training quantization on that target-appliance prediction.

Model definition. Seq2Point is the NILM sequence-to-point modelling approach; the evaluated implementation uses a convolutional neural network (CNN) to estimate the target appliance's power at the centre point of an input sequence. This report refers to it hereafter as the Seq2Point CNN or the evaluated model.

EVALUATION CONFIGURATION



Evaluation framework: MATLAB Deep Learning Toolbox

Quantization Calibration

INT8 post-training quantization was calibrated using 1,000 representative aggregate-input windows of 599 samples each. The calibration windows were sampled across five regions of the evaluation dataset, with 200 windows drawn per region using a fixed random seed (42). The resulting calibration set was used to characterise the activation ranges required for INT8 quantization.

ENGINEERING PROCESS**RVP** · Radianode Validation Process

RVP is Radianode's structured methodology for turning engineering evidence into defensible decisions. It connects the engineering question, expected behaviour, controlled evidence and resulting decision so that conclusions remain traceable to the evidence behind them.

STAGE	DEFINITION	APPLICATION IN ERS-2026-001
RVP-01 Assess	Define the engineering problem by reviewing requirements, understanding system architecture, identifying constraints, and establishing the scope of assessment.	Defined the engineering objective, deployment scenario, and evaluation scope for INT8 post-training quantization of the evaluated Seq2Point CNN.
RVP-02 Simulate	Apply appropriate engineering analysis to evaluate expected system behaviour.	Performed engineering evaluation of the optimised model using representative residential energy data.
RVP-03 Validate	Verify engineering assumptions using objective evidence, performance evaluation, documented findings, and structured engineering review.	Assessed model parameter storage, behaviour preservation, prediction accuracy, and numerical stability through repeatable engineering workflows.
RVP-04 Decide	Evaluate the evidence against requirements, assumptions and remaining uncertainty to establish readiness.	Determined deployment readiness based on the measured engineering evidence presented in this assessment.

01

Model Parameter Storage

Evaluated target: **Water Heater (Boiler)**

ENGINEERING QUESTION

Has model parameter storage been reduced sufficiently to provide a meaningful engineering benefit?

MODEL	MODEL PARAMETER STORAGE
FP32	354.10 KB
INT8	88.52 KB
Measured reduction	75%

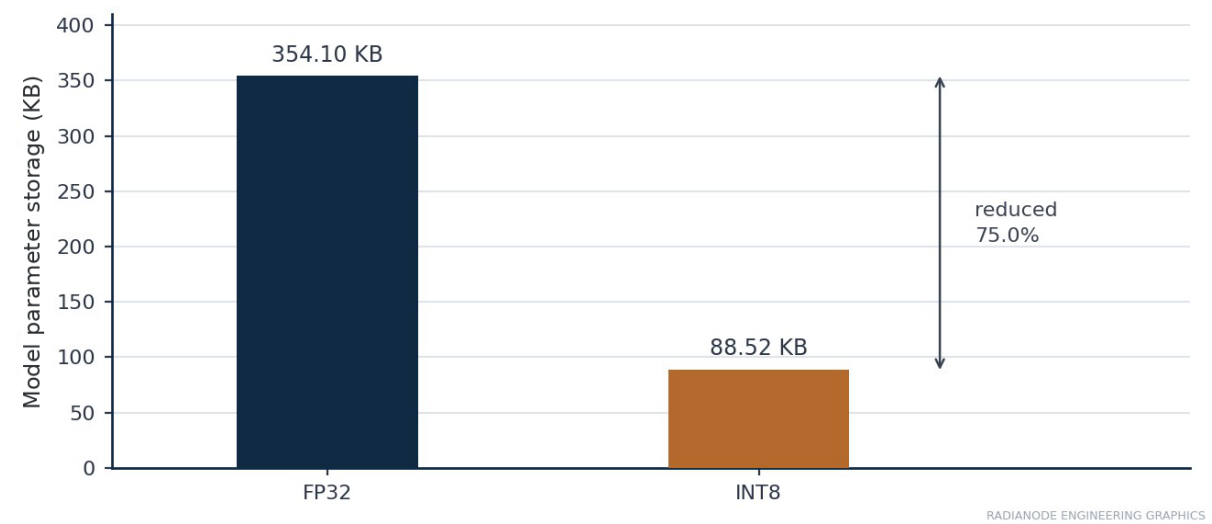
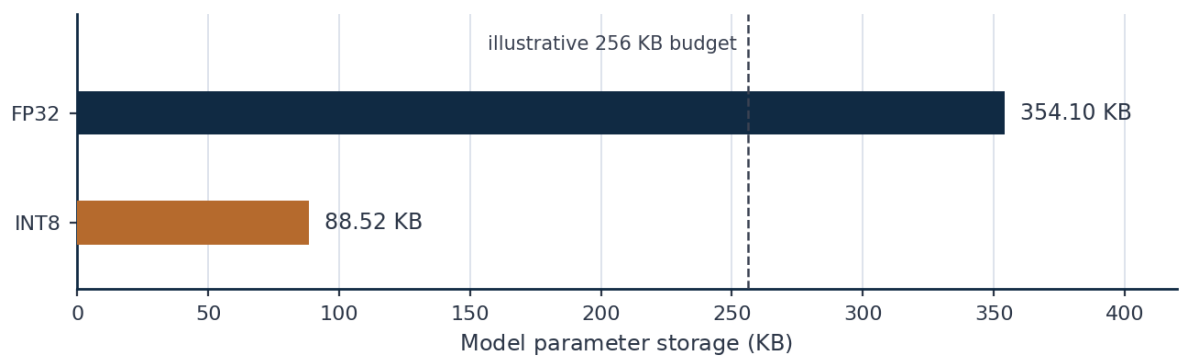


FIGURE 1
Model Parameter Storage Comparison

Model parameter storage for the FP32 and INT8 models. Quantization reduced parameter storage from 354.10 KB to 88.52 KB, a reduction of 75.0%.



RADIANODE ENGINEERING GRAPHICS

FIGURE 2**Model Parameter Storage Breakdown**

The same two parameter-storage values on a linear scale, shown against an illustrative 256 KB platform budget.

Note: The 256 KB platform budget shown in Figure 2 is illustrative and is not representative of a specific production processor or memory architecture.

ENGINEERING INTERPRETATION

Model parameter storage was reduced from 354.10 KB under FP32 representation to 88.52 KB under INT8 representation, corresponding to a 75% reduction in parameter storage. The reduction was achieved without modifying the network architecture or requiring model retraining. This represents a significant improvement in deployment efficiency and increases suitability for memory-constrained embedded platforms.

ENGINEERING FINDING

Model parameter storage was reduced by 75% without modifying the network architecture or requiring retraining.

Behaviour Preservation

ENGINEERING QUESTION

Did the optimised model continue behaving like the original implementation?

METRIC	RESULT
FP32-INT8 prediction correlation	0.9940
Mean absolute FP32-INT8 prediction difference	0.8532 W
Maximum absolute FP32-INT8 prediction difference	2.7062 W

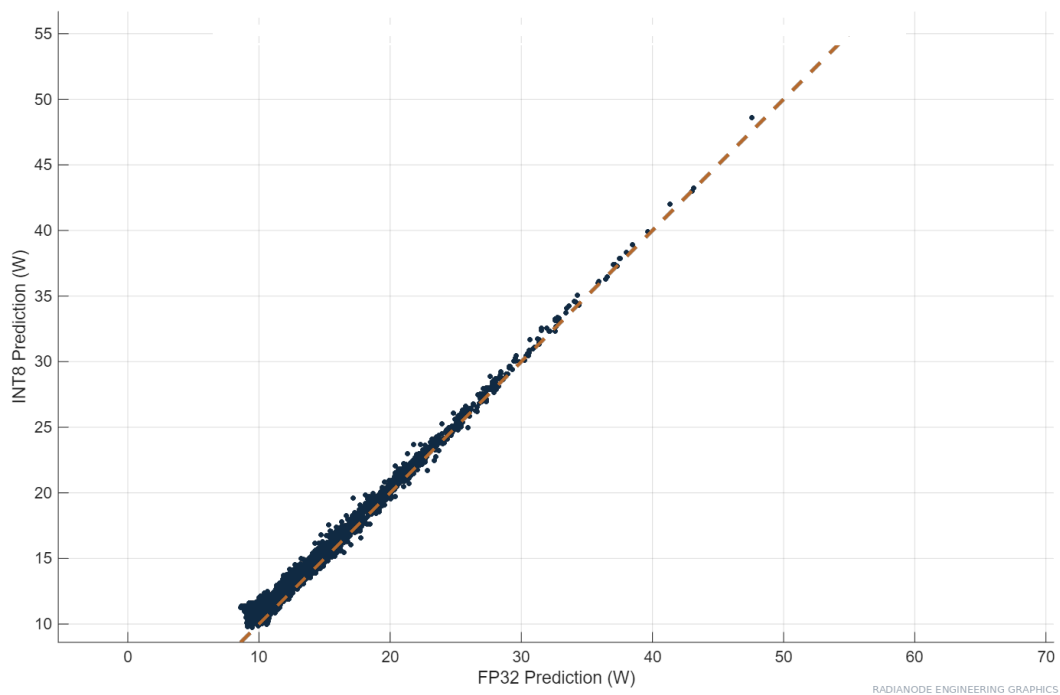


FIGURE 3

FP32 vs INT8 Prediction Scatter

FP32 versus INT8 target-appliance prediction. Each point represents the prediction from the FP32 reference model paired with the corresponding INT8 prediction for the same evaluation sample. The diagonal represents equality between the two implementations.

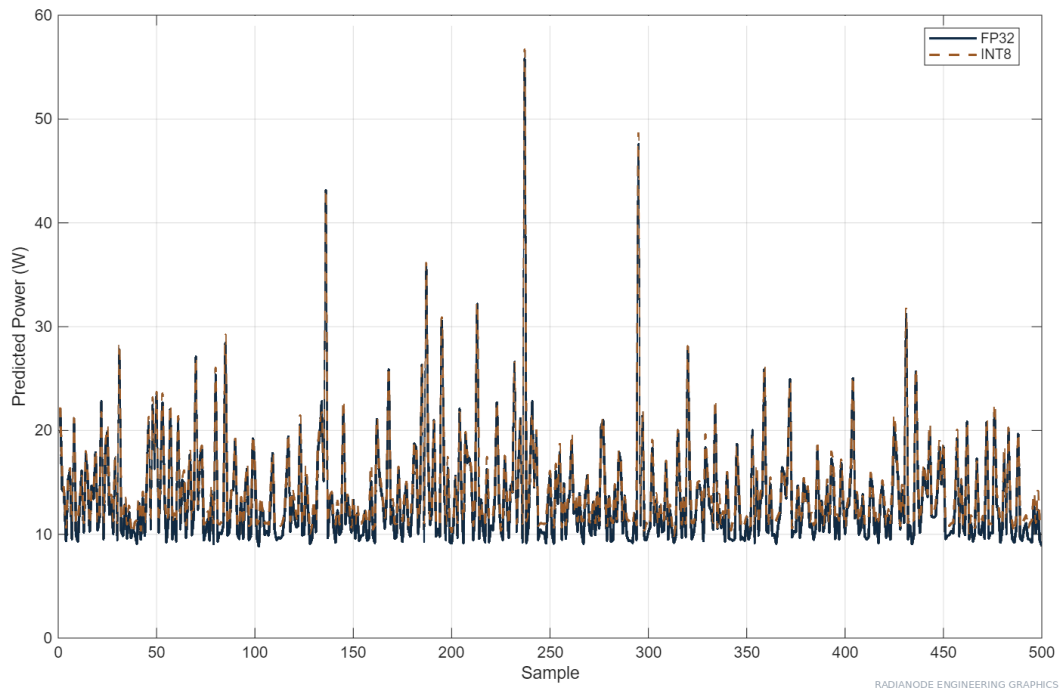


FIGURE 4
Prediction Overlay

FP32 and INT8 predictions of the same target appliance, Water Heater (Boiler), overlaid across the first 500 evaluation samples. The two follow the same transitions, including the isolated high-power events.

ENGINEERING INTERPRETATION

The FP32 and INT8 implementations exhibited a strong linear correspondence across the evaluated samples, with a Pearson correlation of 0.9940. The prediction comparison also identified a small systematic numerical offset, quantified separately through the FP32-to-INT8 error metrics.

ENGINEERING FINDING

Model behaviour remained consistent after optimisation.

Prediction Accuracy

ENGINEERING QUESTION

Was inference performance maintained following optimisation?

Accuracy is evaluated against the corresponding target-appliance ground-truth signal using RMSE and MAE. The comparison is intended to determine whether INT8 quantization materially changes the inference accuracy of the FP32 reference model.

METRIC	FP32	INT8
RMSE	21.6061 W	21.4706 W
MAE	11.7867 W	11.5046 W

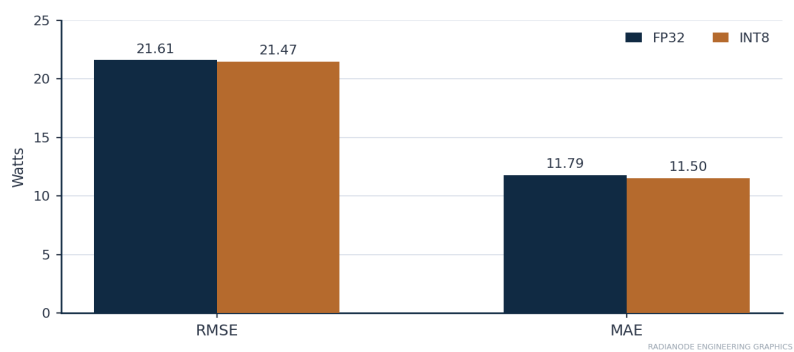


FIGURE 5
Accuracy Comparison

RMSE and MAE of the FP32 and INT8 target-appliance predictions against the corresponding ground-truth signal. The FP32 and INT8 models exhibit closely matched RMSE and MAE values over the evaluated dataset. The slightly lower INT8 RMSE and MAE are not interpreted as an accuracy improvement; the purpose of this comparison is to determine whether quantization materially changes the FP32 reference behaviour.

ENGINEERING INTERPRETATION

Prediction accuracy remained effectively unchanged. The measured differences between FP32 and INT8 are small relative to the absolute error observed in the evaluation and provide no evidence of material degradation caused by quantization.

ENGINEERING FINDING

Prediction accuracy was maintained after optimisation.

Numerical Stability

ENGINEERING QUESTION

Did optimisation introduce deployment-critical numerical effects?

METRIC	RESULT
FP32 to INT8 RMSE	1.0254 W
FP32 to INT8 MAE	0.8532 W
Mean relative error	7.90%
Mean FP32 – INT8 prediction error (offset)	−0.819 W

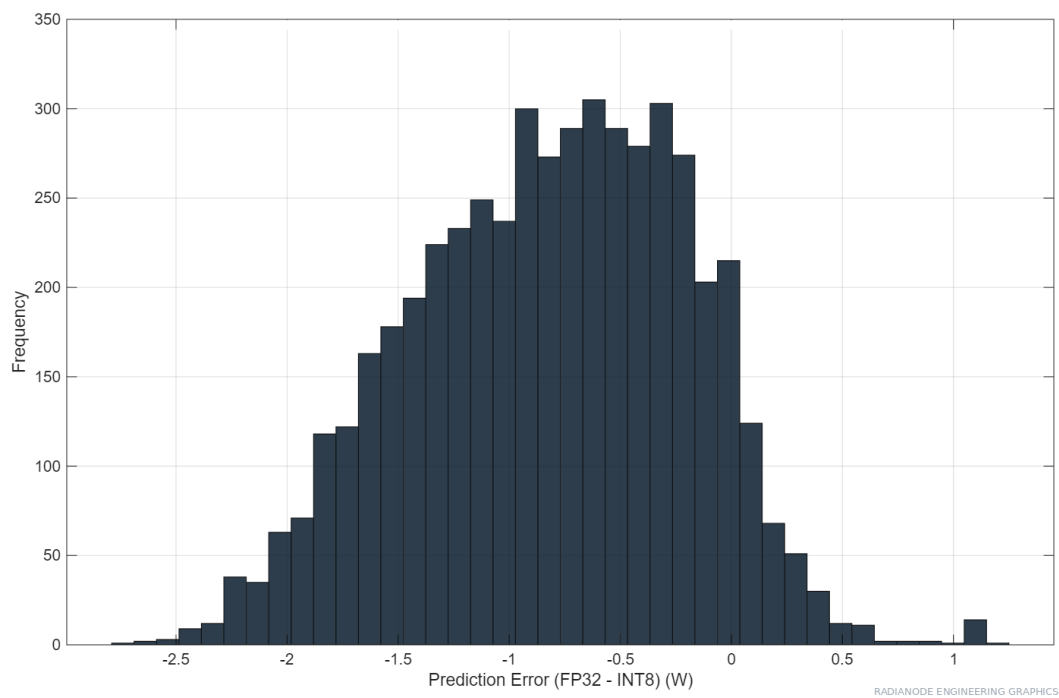


FIGURE 6
Error Distribution Histogram

Distribution of the prediction difference between the two models. The error distribution is concentrated around a negative central offset, with a small isolated positive-side cluster visible in the evaluated samples.

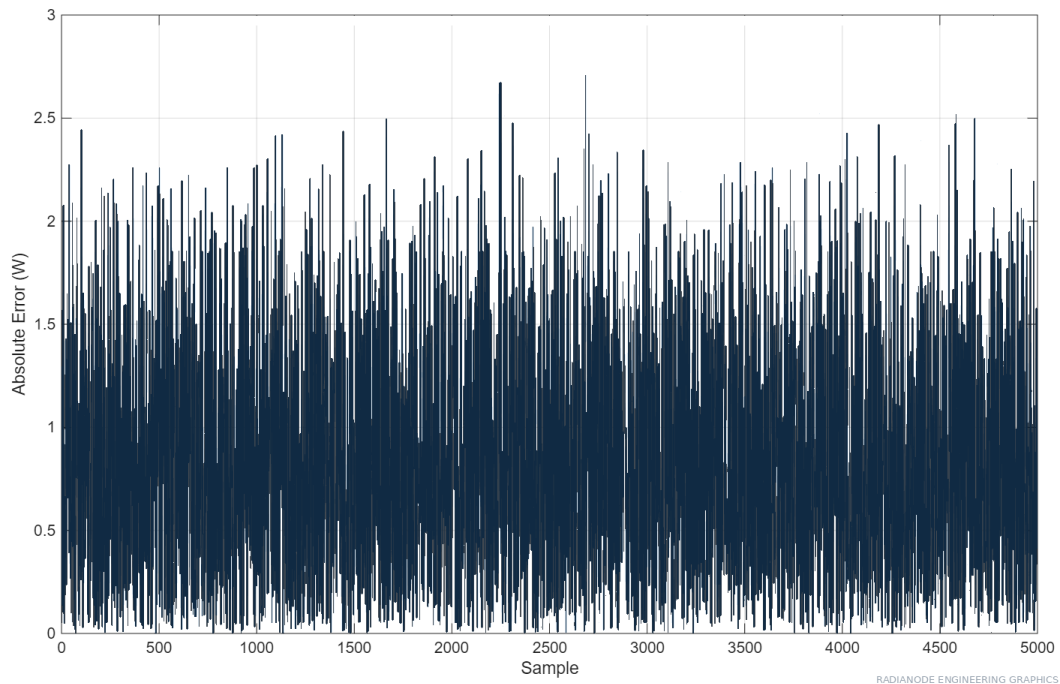


FIGURE 7
Absolute Error

Absolute prediction difference against sample index across 5,000 samples. The plotted error envelope remains broadly stable across the evaluated sequence, with no visible accumulation trend.

ENGINEERING INTERPRETATION

The FP32-to-INT8 comparison exhibits a small systematic prediction offset, while the residual error remains bounded across the evaluated samples.

The mean relative FP32-to-INT8 prediction deviation was 7.90%. This metric reflects numerical deviation between the two implementations rather than degradation relative to ground truth.

The absolute error envelope remained broadly stable across the evaluated sequence, with no visible accumulation trend or evidence of deployment-critical numerical instability.

ENGINEERING FINDING

Numerical deviation remained bounded over the evaluated samples, with no observed evidence of instability introduced by INT8 quantization.

SCOPE OF EVALUATION

Clearly defining the limits of this assessment strengthens the credibility of the engineering conclusions presented.

This assessment establishes

- INT8 reduced the evaluated model's parameter storage by 75%.
- FP32 and INT8 prediction behaviour remained closely aligned.
- No material degradation attributable to quantization was observed in the reported accuracy metrics.
- No instability was observed in the evaluated numerical comparison.

This assessment does not establish

- Real-device execution latency
- Processor utilisation
- Energy consumption
- Hardware-specific optimisation
- Hardware-in-the-loop behaviour
- Cross-dataset generalisation
- Long-term operational performance
- Production-system acceptance

Target-appliance scope: Results apply to the evaluated Water Heater (Boiler) Seq2Point model and should not be interpreted as validation of arbitrary appliance targets.

ENGINEERING EVIDENCE SUMMARY

ENGINEERING CRITERION	EVIDENCE-DERIVED OUTCOME
Model parameter storage	75% reduction
Prediction behaviour	Strong correspondence; $r = 0.9940$
Prediction accuracy	No material degradation observed
Numerical stability	Bounded deviation; no observed instability

RADIANODE ENGINEERING POSITION

Following the engineering evaluation, Radianode considers INT8 post-training quantization to be an appropriate deployment optimisation for the evaluated Water Heater (Boiler) Seq2Point NILM model when deployment objectives prioritise memory efficiency without material degradation in inference performance.

For the evaluated deployment scenario:

- Model parameter storage was reduced by 75%.
- Prediction behaviour remained consistent with the FP32 reference.
- No material degradation in measured inference accuracy was observed relative to the FP32 reference.
- Numerical deviation remained bounded, within the observed numerical behaviour of the evaluation.

Based on the measured evidence, no quantization-related engineering barriers were identified within the evaluated scope.

Deployment validation on the intended production hardware remains recommended as part of final system acceptance.

ENGINEERING READINESS OUTCOME

Ready for edge deployment within the evaluated software-based scope.

This readiness outcome does not constitute production-system acceptance or hardware-level deployment validation.

ENGINEERING CONCLUSION

Within the defined evaluation scope, INT8 post-training quantization preserved the observed prediction behaviour of the evaluated Water Heater (Boiler) Seq2Point CNN while reducing model parameter storage from 354.10 KB to 88.52 KB, corresponding to a 75% reduction.

The FP32 and INT8 implementations exhibited a strong linear correspondence, with a Pearson correlation of 0.9940. The comparison also identified a small systematic numerical offset, while the absolute FP32-to-INT8 prediction error remained bounded over the evaluated samples.

RMSE and MAE against the target-appliance ground-truth signal remained closely matched between the two implementations, with no observed material degradation attributable to INT8 quantization.

These results provide engineering evidence that INT8 post-training quantization can serve as a model parameter-storage optimisation for the evaluated Seq2Point NILM implementation within the defined software-based evaluation conditions.

The conclusion is limited to the evaluated model, dataset and assessment workflow. Validation on intended production hardware, including execution latency, processor utilisation, energy consumption and hardware-specific behaviour, remains outside the present assessment.

REPRODUCIBILITY STATEMENT

This Engineering Readiness Assessment is an implementation of the Radianode Validation Process (RVP).

Supporting engineering artefacts, reproducibility resources, implementation workflows, and publication assets are available through the official Radianode GitHub repository.

Repository contents

- Engineering implementation
- Evaluation workflows
- Figure-generation scripts
- Engineering metrics
- Publication assets
- Reproducibility guide
- Version history

Repository

github.com/radianode/ers-2026-001

APPENDIX A

Figure Traceability

Every published figure is traceable to the engineering workflow that generated it.

REPORT FIGURE	ENGINEERING WORKFLOW	OUTPUT
Figure 1 – Model Parameter Storage Comparison	Memory evaluation	Memory comparison chart
Figure 2 – Model Parameter Storage Breakdown	Memory evaluation	Model parameter-storage comparison chart
Figure 3 – FP32 vs INT8 Prediction Scatter	Behaviour comparison	Prediction fidelity scatter plot
Figure 4 – Prediction Overlay	Behaviour comparison	Prediction overlay plot
Figure 5 – Accuracy Comparison	Accuracy evaluation	RMSE/MAE comparison chart
Figure 6 – Error Distribution Histogram	Numerical stability evaluation	Error distribution histogram
Figure 7 – Absolute Prediction Error	Numerical stability evaluation	Absolute prediction error plot

This appendix strengthens reproducibility by linking each published figure to the engineering workflow that generated it.

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Independent Engineering Firm

This publication (ERS-2026-001) is issued by Radianode Ltd, an independent UK-registered engineering firm. Supporting engineering evidence, implementation artefacts and reproducibility materials are available at the repository referenced in this report's Reproducibility Statement.

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